

Enterprise Cisco / Arista Switch Deployment using Zero Touch Provisioning (ZTP)

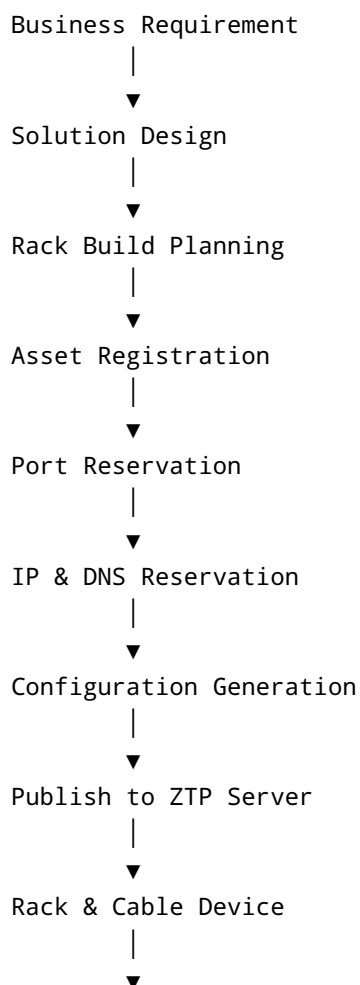
High-Level Deployment Guide

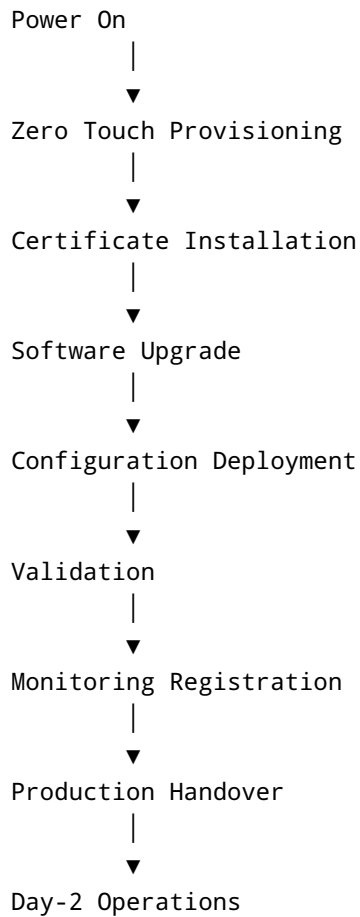
Overview

This document provides a high-level overview of the end-to-end deployment process for Cisco Nexus and Arista EOS switches using Zero Touch Provisioning (ZTP).

The deployment lifecycle starts from infrastructure planning and ends with the switch being handed over to operations for production support.

Enterprise Deployment Flow





1. Business Requirement

A project or application team requests new network infrastructure.

Typical inputs include:

- New rack deployment
- Server expansion
- Kubernetes platform
- VMware cluster
- Storage connectivity
- Disaster recovery environment

Deliverables:

- Project approval
- High-Level Design (HLD)
- Low-Level Design (LLD)

2. Solution Design

The network team prepares the overall design.

This includes:

- Switch models
 - Spine-Leaf topology
 - VLAN allocation
 - Routing design
 - MLAG/vPC design
 - IP addressing
 - Redundancy
 - Software version
-

3. Rack Build Planning

Before hardware arrives, a rack build document is prepared.

Information captured:

- Rack number
- RU position
- Power feeds
- Cable labels
- Patch panel mapping
- Management connectivity

Output:

A complete rack layout used by the data center team.

4. Asset Registration

Every switch is registered in the asset inventory (CMDB).

Typical information:

- Vendor
- Model
- Serial Number
- MAC Address
- Rack location
- Asset ID
- Warranty

This enables lifecycle tracking and future support.

5. Port Reservation

Every physical interface is reserved before installation.

Examples:

- Management interface
- Server-facing ports
- Spine uplinks
- MLAG/vPC peer links
- Storage ports

This prevents cabling errors during deployment.

6. IP & DNS Reservation

The network/IPAM team reserves:

- Management IP
- Loopback IP
- VTEP IP
- Gateway
- DNS entry
- Hostname

These assignments become part of the device inventory.

7. Configuration Generation

Automation generates the switch configuration using templates.

Typical technologies:

- Git
- Ansible
- Jinja2
- NetBox/IPAM
- CMDB

Configuration includes:

- Hostname
- Interfaces

- VLANs
 - Routing
 - Security
 - Logging
 - Monitoring
-

8. Publish to ZTP Server

Generated files are copied to the provisioning server.

The server hosts:

- Bootstrap script
- Device configurations
- Software images
- Root certificates
- Device certificates

The ZTP server becomes the source for initial provisioning.

9. Rack & Cable Device

The data center team performs the physical installation.

Activities include:

- Install rails
- Mount switch
- Connect power
- Connect management cable
- Connect uplinks
- Connect server cables

No manual CLI configuration is required.

10. Zero Touch Provisioning (ZTP)

After power-on, the switch starts ZTP automatically.

Workflow:

1. Obtain management IP from DHCP.
2. Receive the bootstrap script location.
3. Download the bootstrap script.
4. Identify the device using its serial number.

5. Download the correct configuration.
 6. Continue provisioning.
-

11. Certificate Installation

Before secure communication begins, enterprise certificates are installed.

Typical certificates:

- Root CA
- Intermediate CA
- Device certificate (if applicable)

These certificates allow secure communication with:

- Git server
 - Automation platform
 - TACACS+/RADIUS
 - HTTPS APIs
 - Monitoring systems
-

12. Software Upgrade

The bootstrap process checks the running software version.

If required:

- Download approved image
- Install software
- Reboot device
- Resume provisioning

This ensures all devices run standardized software.

13. Configuration Deployment

The final configuration is applied.

Typical settings include:

- Hostname
- Management interface
- Users
- SSH
- AAA

- DNS
- NTP
- Syslog
- SNMP
- VLANs
- Port Channels
- MLAG/vPC
- Routing protocols
- ACLs

Configuration is saved after successful deployment.

14. Validation

Automation verifies the deployment.

Validation includes:

- Device reachable
- SSH working
- Correct software version
- Certificates installed
- DNS resolution
- NTP synchronization
- Routing neighbors established
- Interfaces operational
- Configuration saved

Any failed validation must be corrected before production.

15. Monitoring Registration

The switch is integrated with enterprise monitoring platforms.

Typical integrations:

- Prometheus
- Grafana
- Syslog
- Configuration Backup
- SNMP
- Telemetry

The device is now visible to the operations team.

16. Production Handover

After successful validation:

- CMDB updated
- Change request closed
- Documentation updated
- Monitoring enabled
- Backup scheduled
- Operations team notified

The switch is now considered production-ready.

17. Day-2 Operations

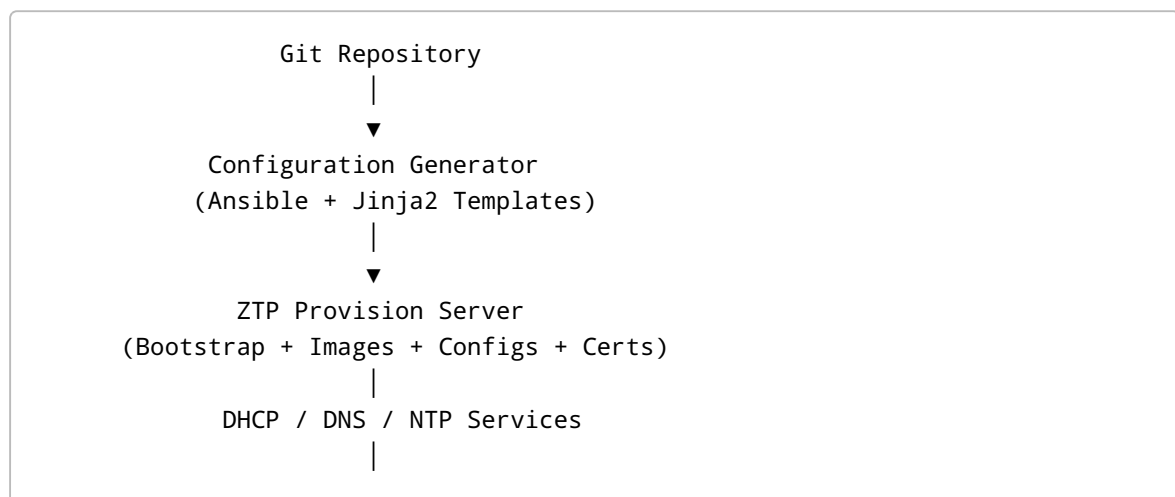
After deployment, the operations team manages the device throughout its lifecycle.

Typical activities include:

- VLAN additions
- Interface changes
- Firmware upgrades
- Certificate renewal
- Configuration backup
- Compliance checks
- Performance monitoring
- Hardware replacement (RMA)
- Troubleshooting

Automation tools are commonly used to perform these tasks consistently.

High-Level Architecture




```
Management Network
|
Cisco / Arista Switch
|
Spine / Core Network
|
Servers / Storage / Kubernetes / VMware
```

Deployment Summary

Phase	Purpose
Planning	Define network design and capacity
Rack Build	Prepare rack layout and cabling
Asset Registration	Register switch information
Port Reservation	Reserve physical interfaces
IP Reservation	Allocate IPs and hostnames
Configuration Generation	Create device configuration
ZTP	Automated provisioning
Certificate Installation	Enable secure enterprise communication
Software Upgrade	Standardize operating system version
Configuration Deployment	Apply production configuration
Validation	Verify successful deployment
Monitoring	Register with monitoring platforms
Production Handover	Transition to operations
Day-2 Operations	Ongoing lifecycle management